

B2 - BACnet / Modbus (Advanced)

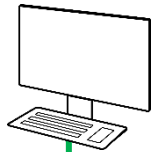
BACnet MS/TP - Workshop I

6-way EPIV4

B2 - BACnet - Workshop I

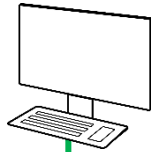
6-way EPIV4

Team A
YABE



Addr = ____
InstNr = ____

Team B
YABE



Addr = ____
InstNr = ____

First define the network addresses within your group

Group 1: BACnet Addresses 10..19, Instance IDs 110..119

Group 2: BACnet Addresses 20..29, Instance IDs 120..129

Group 3: BACnet Addresses 30..39, Instance IDs 130..139

Group 4: BACnet Addresses 40..49, Instance IDs 140..149

Baudrate 19'200
Max Master 127



BKN
Addr = ____
InstNr = ____



PRCA-BAC
Addr = ____
InstNr = ____



22DTH-16M
Addr = ____
InstNr = ____



EP015R6+BAC-HHM
Addr = ____
InstNr = ____



EP015R2+BAC
Addr = ____
InstNr = ____



P-22RTM-1U00D-2
Addr = ____
InstNr = ____

Serial network (RS485)

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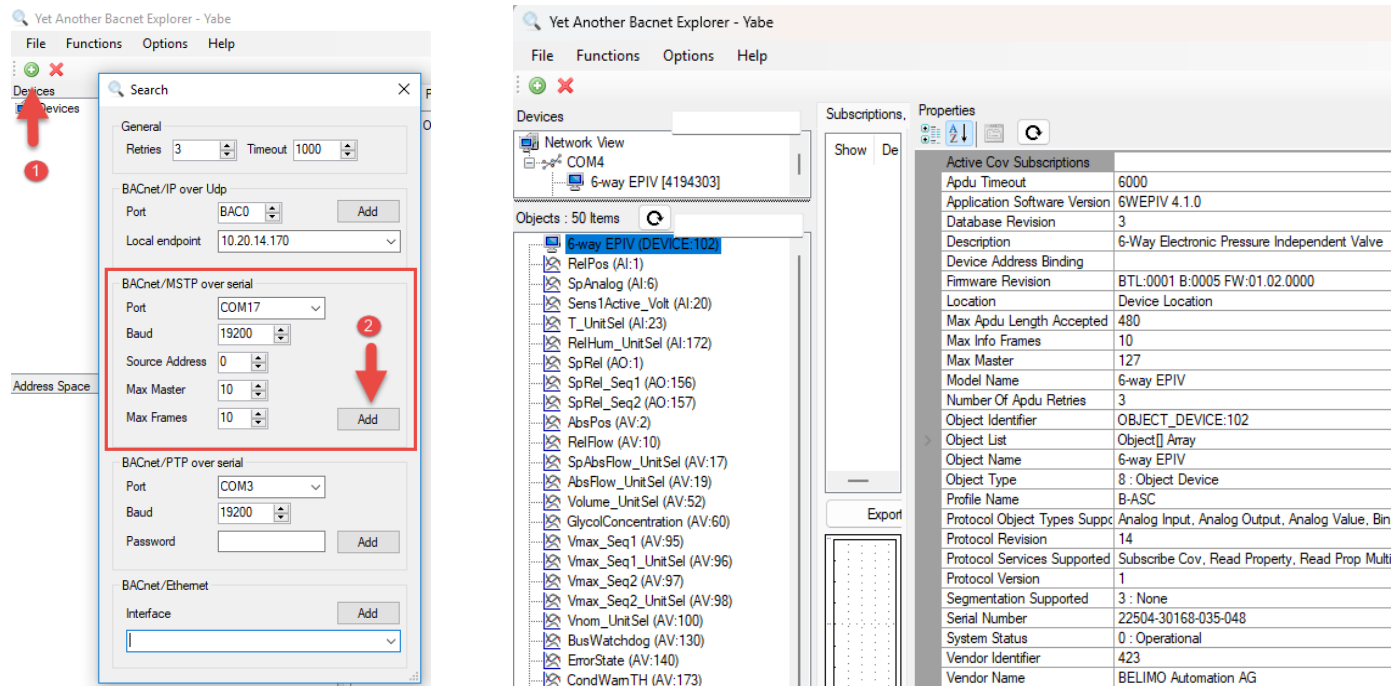
1. Configure all the devices according the network overview.

💡 Find your appropriate configuration tool in the documents (LINK.10, Belimo Assistant 2, etc.)

2. Start YABE. Click on **Add device** and **Add yourself with BACnet/MSTP**.

💡 Check on which COM port your RS485-USB adapter is

💡 Pay attention to the BACnet settings on your actuator you just made



The image displays two screenshots of the 'Yet Another Bacnet Explorer - Yabe' software interface.

The left screenshot shows the 'Search' dialog box. A red arrow labeled '1' points to the 'Add' button in the 'BACnet/MSTP over serial' section. A red arrow labeled '2' points to the 'COM17' port selection in the same section.

The right screenshot shows the 'Properties' window for the '6-way EPIV (DEVICE:102)'. The 'Active Cov Subscriptions' table is visible, showing various parameters for the device.

Active Cov Subscriptions	
Adu Timeout	6000
Application Software Version	6WEPIV 4.1.0
Database Revision	3
Description	6-Way Electronic Pressure Independent Valve
Device Address Binding	
Firmware Revision	BTL:0001 B:0005 FW:01.02.0000
Location	Device Location
Max Adu Length Accepted	480
Max Info Frames	10
Max Master	127
Model Name	6-way EPIV
Number Of Adu Retries	3
Object Identifier	OBJECT_DEVICE:102
Object List	Object[] Array
Object Name	6-way EPIV
Object Type	8 : Object Device
Profile Name	B-ASC
Protocol Object Types Supp	Analog Input, Analog Output, Analog Value, Bin
Protocol Revision	14
Protocol Services Supported	Subscribe Cov, Read Property, Read Prop Multi
Protocol Version	1
Segmentation Supported	3 : None
Serial Number	22504-30168-035-048
System Status	0 : Operational
Vendor Identifier	423
Vendor Name	BELIMO Automation AG

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6-way EPIV4

3. Select the 6-way EPIV4 under *Devices*.
4. Read out all the BACnet objects of the device.
5. Click on the BACnet object “Device”
 1. What is the *Application Software Version*?
 2. With which BACnet object(s) can you write the setpoint(s)?
 3. With which BACnet object can you read out the relative position?
 1. _____
 2. _____
 3. _____
6. Try as many COV subscriptions as possible. How many COV subscriptions could you make?

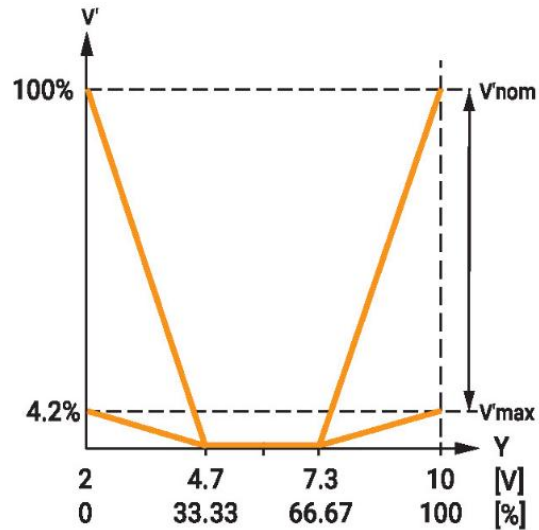
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7. Please check out the *mode of operation* in the datasheet.

1. Which setpoint range is within Sequence 1?

2. Which setpoint range is within Sequence 2?



1.

2.

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8. Which *Present Value* of “SpRel” I have to write in order to get Vmax of sequence 1?

9. Which *Present Value* of “SpRel” I have to write in order to get Vmax of sequence 2?

10. Write the smallest possible “Vmax1” and “Vmax2”.

1. What is the minimal Vmax?

2. What happens if I write a smaller value?

1.

2.

11. Set “Vmax1” and “Vmax2” back to 100%.

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- 12. Set Setpoint source to "Bus" and control mode to "Position control".**
Set Setpoint mode BV[155] to "Separate setpoints".
Write 50 % to Setpoint Sequence1 AO[156].
Observe relative position AI[1] on trend using COV subscription.

Actual relative position value:

Write also 50 % to Setpoint Sequence2 AO[157].
Observe relative position AI[1] on trend using COV subscription.

Actual relative position value:

Read error state AV[140]

Value:

Interpretation:

Set setpoint mode BV[155] back to "Single setpoint".

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13. Set a setpoint AO[1] of 100%.

Observe relative position AI[1] change.

Set condensation protection action MV[171] to "Close Valve" .

Read relative humidity AI[172].

Set Condensation alarm threshold AV[174] to a lower value than relative humidity, e.g., *Rel. Humidity: 40% , Alarm Threshold: 35 %* .

Observe relative position AI[1] change.

Read condensation protection status MV[108].

Value:

Interpretation:

Set Condensation alarm threshold AV[174] to a higher value than relative humidity, e.g., *90%*.

Disconnect the Wago terminal from humidity sensor.

Read condensation protection status MV[108].

Value:

Interpretation:
